

Abhay Manhas

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PROFESSIONAL SUMMARY

Python Backend Engineer with **3+ years** of experience building scalable, production-grade systems using Django and Django REST Framework. Strong expertise in asynchronous programming, distributed task processing, and cloud-native deployments. Hands-on experience designing AI-powered backend systems including LLM orchestration, RAG pipelines, and agent-based workflows. Actively deepening AI/ML knowledge with practical implementation using PyTorch and modern NLP tooling. Seeking backend or AI-platform roles in large-scale engineering environments.

PROFESSIONAL EXPERIENCE

Senior Python Developer

April 2023 – Present

Shubpy Pvt. Ltd.

Chandigarh

- Architected a hybrid-cloud AI orchestration pipeline using OpenRouter to dynamically route workloads across Gemini, Azure OpenAI, and Qwen, reducing AI task costs by approximately 35%.
- Designed and implemented an agent-based Retrieval-Augmented Generation (RAG) system using MCP connectors to trigger operational workflows, achieving 0.85 F1 score on internal retrieval benchmarks.
- Built a MapReduce-style distributed document analysis service using Celery and RabbitMQ, enabling summarization of 500+ page documents in under one minute through task parallelization.
- Deployed and managed applications on Azure Kubernetes Service with auto-scaling to handle variable traffic loads.
- Implemented CI/CD pipelines using Azure DevOps and improved deployment reliability, reducing production incidents by 90%.

Junior Python developer.

June 2022 - Dec 2022

CNT Technologies

Chandigarh

- Developed a real-time chat application using asynchronous programming and WebSockets, supporting 100+ concurrent users with message latency under 100ms
- Migrated legacy **monolithic** Python applications into **Microservices architecture**.Refactored unstructured codebases into a clean three-layer architecture, significantly improving maintainability and readability..

PROJECTS

AILYZE (Ai Qualitative Analysis Tool)

www.ailyze.com

Developed the backend using modular architecture, enforcing separation between business logic and data access layers.Reduced debugging time and long-term maintenance overhead by approximately 60% through improved code organization.

Food Ordering Website

www.chds.com.au

Engineered and deployed a high-performance backend capable of sustaining 10K+ concurrent requests with minimal latency.Optimized database connection pooling and query handling to maintain performance during peak traffic.

TECHNICAL SKILLS

Languages & Frameworks :Python, Django, Django REST Framework (DRF), Channels, WebSockets, Redis, Asyncio, Celery, Pandas, Numpy.

DevOps & Cloud Infra :AWS, GCP, Azure, Docker, Azure Kubernetes Service (AKS), CI/CD, Git, Azure Functions, Blob Storage, Azure Database for PostgreSQL.

AI/ML System & Integrations :PyTorch, Scikit-learn, Hugging Face Transformers, Model Training & Evaluation, Crewai, Gemini, OpenAI, OpenRouter, Claude, RAG, NLP, MCP Server.

Tools& Platforms :Zapier, n8n, Prometheus, Grafana, Flower, Pytest, Unittest, Stripe.

CERTIFICATIONS & ACHIEVEMENTS

Python Certified, HackerRank [Link](#) || CrewAI Certified - Multi-agent artificial intelligence systems [Link](#) || KCNA — Kubernetes & Cloud Native Associate [Link](#) | Forage Academy Data Labeling Job Simulation [Link](#) || Completed 25-hour PyTorch Deep Learning Course [Link](#) ||

EDUCATION

Gura Nanak dev university

Bachelor Of Computer Application

Amritsar

Aug. 2019 – May 2022